

While Loop Practice Exercises

Exercise 1 — Print Numbers 1 to 10

Write a Java program that uses a `while` loop to print the numbers from **1 to 10**.

Expected Output:

```
1
2
3
4
5
6
7
8
9
10
```

Exercise 2 — Print Numbers 10 to 1

Write a Java program that uses a `while` loop to print the numbers from **10 down to 1**.

Expected Output:

```
10
9
8
7
6
5
4
3
2
1
```

Exercise 3 — Print a String in Reverse Order

Write a Java program that creates a `String` and uses a `while` loop to print the characters in **reverse order**.

Example:

```
Input: Java
Output: avaJ
```

Hint:

- Use `length()` to find the length of the `String`.
 - Use `charAt()` to access each character.
 - Start from the last character and move toward the first character.
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Exercise 4 — Print a String but Exclude a Character

Write a Java program that reads a String from the user and prints the String **without a specified character**.

Example:

Input: Java
Character to exclude: a

Output: Jv

Exercise 5 — Calculate the Sum from 1 to 100

Write a Java program that uses a `while` loop to calculate and print the sum of the numbers from **1 to 100**.

Expected Output:

Sum = 5050

Exercise 6 — Positive Number

Write a Java program that asks the user to enter a number.

Keep asking the user to enter a number **while the number is less than or equal to 0**.

When the user enters a positive number, print:

Valid number

Example:

Enter a number: -5
Enter a number: 0
Enter a number: -10
Enter a number: 8
Valid number

Exercise 7 — Break and Continue with String

Write a Java program that repeatedly asks the user to enter a word.

Use a `while` loop.

- If the user enters "skip", use `continue` to **skip that word** and ask for another word.
- If the user enters "quit", use `break` to **stop the loop**.
- Otherwise, print the word entered by the user.
- The comparison should be **case-insensitive**.

Example

Enter a word: Java
Java

Enter a word: skip

Enter a word: String
String

Enter a word: SKIP

Enter a word: Programming
Programming

Enter a word: QUIT
Program stopped.

Requirements

- Use `while`
 - Use `break`
 - Use `continue`
 - Use `equalsIgnoreCase()`
 - Do not use a `for` loop
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Exercise 8 — Caesar Cipher (Challenge)

Write a Java program that encrypts a `String` by shifting each letter **3 positions forward** in the alphabet.

For example:

A → D
B → E
C → F

Use a `while` loop to process each character.

Expected Output

Input: PROGRAMMING
Output: SURJUDPPLQJ

Hints

- Use `charAt()` to access each character.
- Use `length()` to determine how many characters need to be processed.
- Use `% 26` to wrap around after `z`.
- Use a `while` loop to process each character.
- Assume the input contains uppercase letters only.

Example of Wrapping

X → A
Y → B
Z → C